Build the Recommender

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Teach the code together with its assumptions and tests.

Run all supplied code, keep policies separate and assess the selected mystery-book case before feedback.

Keep one habit: Require rejected versus ineligible counts and a tie explanation tied to the current task.

MAIN LESSON · SECTION 1

Prepare the main lesson and its retained extension

> Use fictional records and schedule separate extension time.

Goal and preparation

Primary target: Implement a **deterministic recommender**

Teach Week 24 as a two-session investigation using fictional data. Build a fictional activity **recommender** using last week's score. Require duration at most a supplied limit. On a tie choose the alphabetically first name.

This makes the outcome reproducible rather than dependent on **input** order. Assume unique names and validated numeric fields. Assess the three stated outcomes through the independent task, not by copying the worked result.

By the end, I can implement **deterministic** filtering, scoring and ties for a complete catalogue case.

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Main plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval and readiness

Without opening the old lesson, explain one key idea from Project Problem and Success Test. Give an example and a mistake that the idea helps you avoid.

Look for: Define a small application purpose

Reference example, not the only acceptable wording: Suggest a suitable eligible activity for a teacher-supervised session; the teacher reviews and decides.

Without opening the old lesson, explain one key idea from Agents and Tools. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of state machines for bounded agents

Reference example, not the only acceptable wording: Teacher **explanation**: A bounded agent uses explicit states and finite action limits. A state machine makes invalid transitions visible.

States are ready, looking_up, review, done. ready may start lookup; lookup may finish done or request review; review may finish done after authorised **evidence**. done is terminal. Maximum two lookup attempts.

Without opening the old lesson, explain one key idea from Predictive vs Generative. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish discriminative **prediction** from generation by inspecting the requested **output**

Reference example, not the only acceptable wording: Teacher **explanation**: Message labels are discriminative **prediction**; continuing a sequence is generation; record lookup is retrieval rather than learned generation by itself.

One application can route a request, retrieve **evidence** and generate a response.

No. Filter by the hard constraint before ranking.

▼ Materials and extension planning

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional **input** records from this lesson; no real student identifiers

Apply the connected idea: Building the data pipeline

Teach Week 32 as a two-session investigation using fictional data. Build the validation **stage** of the planned chooser first. A valid record has nonempty text name, integer minutes 0 through 60 and integer fun and learning ratings 0 through 5. The batch **stage** must separately check name uniqueness. Only accepted records may reach the selection function in this week's lesson. Assess the three stated outcomes through the independent task, not by copying the worked result.

Readiness answer

Validate, filter eligibility, calculate scores, resolve ties, then report with limits.

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional **input** records from this lesson; no real student identifiers

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Case exposure check: this source example appears in ai-2/24, ai-2/33. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Project Problem and Success Test

Agents and Tools

Predictive vs Generative

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Catalogue: A mystery/90/illustrated, B mystery/140/not illustrated, C science/80/illustrated. Request mystery, maximum 100 pages. Rank with 2 genre points and 1 illustration point after filtering.

Show why other books are excluded.

Concept to explain

A transparent **baseline** makes later improvements understandable.

Term	Meaning
baseline	A simple reference method used to judge whether a more complex method adds value.
candidate	An item or proposed answer being considered before a final selection.
weighted total	A sum formed after multiplying each contribution by its assigned weight.
explanation	An account connecting a result to the mechanism, calculation or evidence that supports it.

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; **explanation** 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Model the activity state trace and separate book ranking

> Keep schemas, scores and tie rules labelled.

State main assumptions

Build a fictional activity **recommender** using last week's score. Require duration at most a supplied limit. On a tie choose the alphabetically first name.

This makes the outcome reproducible rather than dependent on **input** order. Assume unique names and validated numeric fields.

Run and trace the activity selector

```
def recommend(items, limit):
    best = None
    best_score = None
    for item in items:
        if item["minutes"] <= limit:
            score = 2 * item["fun"] + item["learning"]
            if (best is None or score > best_score
                or (score == best_score and item["name"] < best)):
                best = item["name"]
                best_score = score
    return best

items = [{"name": "B", "fun": 3, "learning": 2, "minutes": 10},
```

```
          {"name": "A", "fun": 2, "learning": 4, "minutes": 9}]
print(recommend(items, 10))
```

The function prints nothing; the caller prints A. best starts as None to mean no eligible option found. Python short-circuits or: when best is None, later comparisons in that condition need not run.

After selecting an item, best_score is numeric. An empty list or no eligible item returns None.

> **Predict the result if the list order is reversed, and if the limit is 8.**

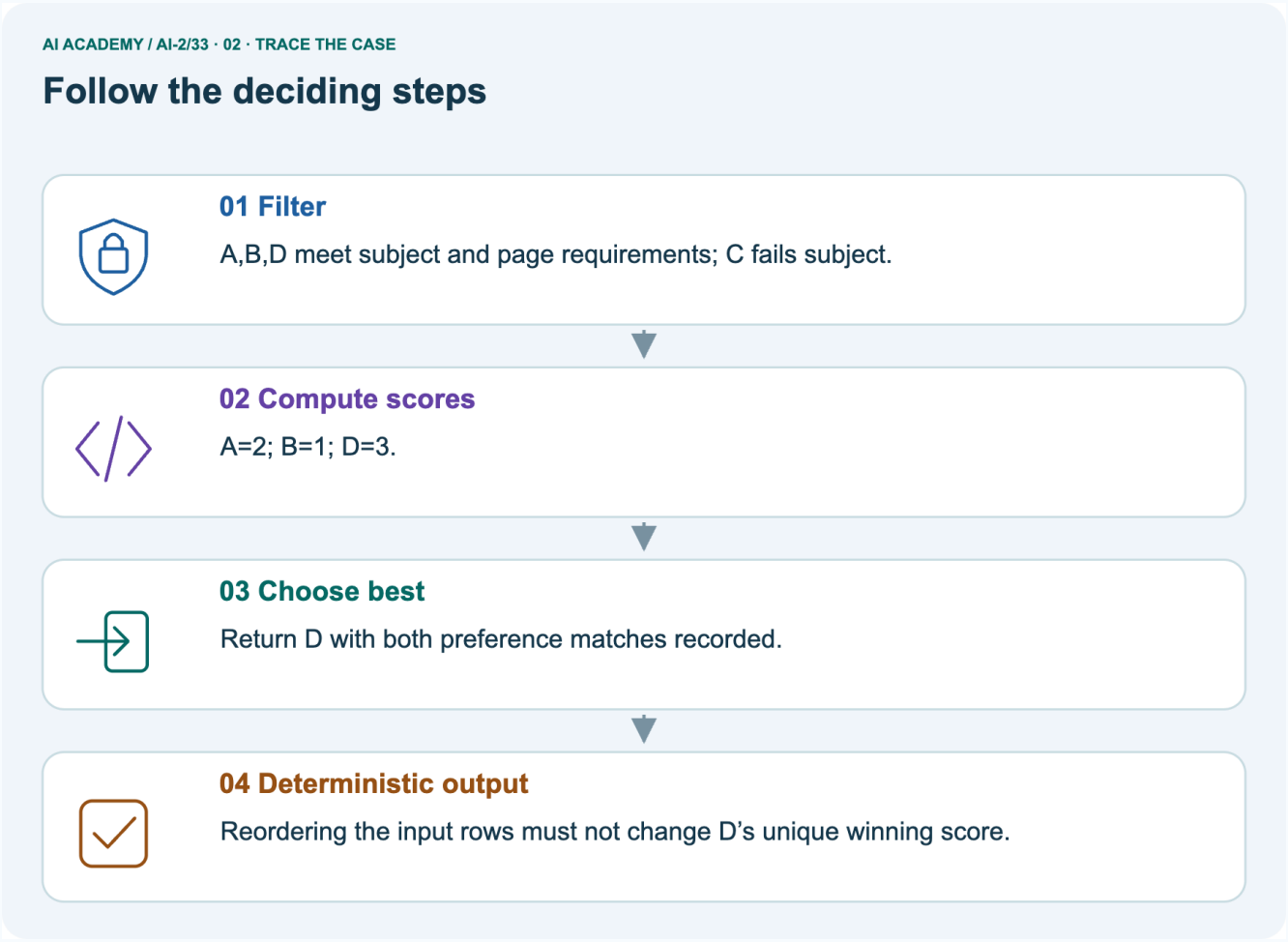
Read the separate animal-book dataset

A fictional book chooser filters a request before comparing preference scores.

Request: animal subject, at most 90 pages. Score 2 for illustrated and 1 for comic. Tie chooses smaller ID.

A animal90 illustrated prose; B animal70 plain comic; C space60 illustrated comic; D animal80 illustrated comic.

Trace the book ranking and exclusions



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Filter	A,B,D meet subject and page requirements; C fails subject.
2. Compute scores	A=2; B=1; D=3.
3. Choose best	Return D with both preference matches recorded.
4. Deterministic output	Reordering the input rows must not change D’s unique winning score.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Eligibility and ranking answer different questions. C has a high preference score but is not an animal book, so it must stay excluded.

The implementation needs a no-match result and explicit tie policy even when this example has a unique winner. Save rejected reasons for an **explanation**, not just the final ID.

Predict the changed-D result

Change: Change D’s pages to 91, leaving every other field unchanged.

Prediction: D becomes ineligible, so A wins with score 2.

Reason: The strict excess over the 90-page maximum changes filtering before ranking.

Run the complete pictured book code

```
def choose_book(rows, subject, max_pages):
    # Inputs are supplied, validated fictional rows with unique IDs.
    eligible = [r for r in rows if r['subject'] == subject and r['pages'] <= max_pages]
    ranked = sorted(eligible, key=lambda r: (-(2*int(r['illustrated']) + int(r['comic'])),
r['id']))
    return [(r['id'], 2*int(r['illustrated']) + int(r['comic'])) for r in ranked]

books = [dict(id='A',subject='animal',pages=90,illustrated=True,comic=False),
         dict(id='B',subject='animal',pages=70,illustrated=False,comic=True),
         dict(id='C',subject='space',pages=60,illustrated=True,comic=True),
         dict(id='D',subject='animal',pages=80,illustrated=True,comic=True)]
print(choose_book(books, 'animal', 90))
changed = [dict(r, pages=91) if r['id']=='D' else dict(r) for r in books]
print(choose_book(changed, 'animal', 90))
print(choose_book(books, 'animal', 50))
```

Explain the picture's limits

Toy preference weights do not imply shorter or illustrated books are universally better for all learners.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Pipeline extension · Validate each row before selection

Build the validation **stage** of the planned chooser first. A valid record has nonempty text name, integer minutes 0 through 60 and integer fun and learning ratings 0 through 5. The batch **stage** must separately check name uniqueness. Only accepted records may reach the selection function in this week's lesson.

```
def valid_item(row):
    if type(row) is not dict:
        return False
    if type(row.get("name")) is not str or not row["name"]:
        return False
    for key, maximum in [("minutes", 60), ("fun", 5), ("learning", 5)]:
```

```
            value = row.get(key)
            if type(value) is not int or not 0 <= value <= maximum:
                return False
        return True

    print(valid_item({"name": "A", "minutes": 20, "fun": 4, "learning": 2}))
    print(valid_item({"name": "B", "minutes": 10, "fun": "4", "learning": 2}))
```

The worked calls print True then False. Short-circuit checks keep absent or wrongly typed fields away from later operations.

This Boolean helper does not explain the reason; add reason codes in the reporting **stage** or use a validation result record. Do not describe a range failure as a no-match outcome.

The first passes. Replacing minutes with 61 fails the range. Exit: No. The batch uniqueness contract fails. Exclude every occurrence of each duplicated name and report the reasons for correction, following the frozen Week32 policy.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Input → **Process** → **Output**

Trace the information. At each **stage**, ask what changed and how you could test it.

Which **stage** could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Weights: genre=3, difficulty=2, length=1.

2. Compute every **candidate**'s terms and total.

3. Filter hard constraints, sort remaining totals high to low.

4. **Output** top choice, score breakdown, one limitation, and a tie rule.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

A has flags (1,1), B=(1,0), C=(0,1); score=2g+l.

Compute the winner and **evidence** to display.

Exact explanation to model

A wins at 3; B=2; C=1. Show flags, formula, scores and **explanation**; declare tie policy.

Explain the added worked example

Key terms

Term	Meaning
Recommender	A procedure that suggests options using a stated policy
Tie	Two options have the same comparison score
Deterministic	The same inputs and rules produce the same result
Explanation	Evidence describing how a result was obtained

Apply the connected idea: Building the data pipeline

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal **explanation** together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Build the Recommender

The facts we will use

AI ACADEMY / AI-2/33

Filtering before scoring

	Pages	Preference score	Status
A	90	2	eligible
B	70	1	eligible
C	60	3	subject fails
D	80	3	eligible

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/33

Build the Recommender

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Filter

A,B,D meet subject and page requirements; C fails subject.

2. Compute scores

A=2; B=1; D=3.

3. Choose best

Return D with both preference matches recorded.

4. Deterministic output

Reordering the input rows must not change D's unique winning score.

CHANGE ONE THING

Change D's pages to 91, leaving every other field unchanged.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

MAIN LESSON · SECTION 3

Guide the main explanation, then fade support

> Collect each learner’s reasoning before revealing a key.

Main completion question and key

Completion task: State recommend([], 10) and the worked items with limit 8.

Teacher check: Both return None; the caller should display no eligible option.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Main individual check

Use the worked case for Implementing a **recommender**. Proposed **explanation**: “A successful run proves the conclusion”. Decide whether this **explanation** is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: The recommendation follows a written scoring policy. It does not learn preferences, prove learning benefit or authorize an action.

Display the chosen rule and let a person review whether an option is suitable.

Collect **explanation**s, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group **explanation** while others solve a more demanding variation.

Pictorial question and changed case

Explain why B does not win after D is excluded even though it has fewer pages.

Page count is an eligibility limit, not a ranking preference here. A scores 2 and B scores 1.

Changed-case reasoning: D becomes ineligible, so A wins with score 2. The strict excess over the 90-page maximum changes filtering before ranking.

Collect each learner's **explanation** before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep learners explaining

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an **explanation**.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Extension teaching cycle

Target: Separate validation selection and reporting

Reserve further sessions for this topic; the 135-minute primary plan does not absorb all added content.

Completion practice: What does valid_item return if learning is absent?

Teacher check: False: get returns None and the exact integer check fails.

Individual check: For Building the data pipeline, use its worked case to evaluate this proposed **explanation**: "A successful run proves the conclusion". Explain the deciding step.

Expected correction: Each **stage** should have a clear **input** and **output** contract. A program can be correct at one **stage** but wrong at their connection, such as passing rejected rows into selection.

Keep acceptance and rejection counts in the report.

Apply the same support, independent transfer and delayed-review decisions to the named connected-topic workbook tasks.

▼ Earlier wording, alternatives and picture transcripts

Demonstrate and guide practice

Board sequence

Write the given facts: A has flags (1,1), B=(1,0), C=(0,1); score=2g+l.

Write the question to explain: Compute the winner and **evidence** to display.

Reveal after **prediction**s: A wins at 3; B=2; C=1. Show flags, formula, scores and **explanation**; declare tie policy.

Use the earlier transfer question as optional practice. Reserve Rank the new mystery catalogue independently for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Reversing still returns A. Limit 8 leaves no eligible item and returns None. Exit: X meets the 12-minute limit and ties for highest eligible score 9; the alphabetical tie rule selects X.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and explain the purpose in the opening section.
7 to 20 minutes	Read the worked example and diagram; ask learners to trace each step aloud.
20 to 33 minutes	Complete W1 and W2, then compare predictions with observed results.
33 to 45 minutes	Collect the guided prediction before running; discuss a mismatch and its cause.
Session 2 start to 8 minutes	Retrieve the main rule without showing the previous answer.
8 to 20 minutes	Use W3 to diagnose and repair the stated failure; rerun an earlier case.

Stage	Teacher action
20 to 35 minutes	Collect W4 independently, including code or plan, predictions and explanations, before hints.
35 to 45 minutes	Discuss W5 and the exit; use the fresh retry when support was needed.

Teach the connected idea

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and explain the purpose in the opening section.
7 to 20 minutes	Read the worked example and diagram; ask learners to trace each step aloud.
20 to 33 minutes	Complete W1 and W2, then compare predictions with observed results.
33 to 45 minutes	Collect the guided prediction before running; discuss a mismatch and its cause.
Session 2 start to 8 minutes	Retrieve the main rule without showing the previous answer.
8 to 20 minutes	Use W3 to diagnose and repair the stated failure; rerun an earlier case.
20 to 35 minutes	Collect W4 independently, including code or plan, predictions and explanations, before hints.
35 to 45 minutes	Discuss W5 and the exit; use the fresh retry when support was needed.

Reduce help deliberately

Individual reasoning check

Connected teaching cycle: Building the data pipeline

Pictorial practice question

Reasoned teacher answer

Run the supplied main investigation

> Keep the selected independent task out of shared practice.

Supplied records and answer guidance

Activity example: A and B score 8; A wins by name in either order; limit 8 returns None. Complete pipeline: A, B and C validate; D fails the fun type; B is valid but too long; A and C tie at 8 and A wins. Reversing input preserves ranking; zero limit gives no match; Boolean True is an invalid request.

Reject every duplicated A occurrence, leaving C eligible. Separate book picture: A scores 2, B 1, D 3; C fails subject. Changing D to 91 pages selects A. Independent mystery task: D fails length, F fails subject; E scores 3, G and H score 5; H wins with 90 rather than 95 pages. X/Y/Z remains additional activity practice.

Predict, investigate and record

1. Predict → Test → Record → Explain

2. Keep expected and actual results separate. Surprises are useful data when recorded honestly.

3. Run the supplied complete activity pipeline and pictured book ranking on fictional local records. Hand-check scores, reverse order, test duplicates and invalid fields, and compare a valid no-match result with an invalid request. No live learner profile is required.

4. What did the hardest test teach you that the normal test did not?

Keep the evidence

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run

Record	What to capture
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Main misconceptions

A successful run proves the conclusion: The recommendation follows a written scoring policy. It does not learn preferences, prove learning benefit or authorize an action.

Display the chosen rule and let a person review whether an option is suitable.

Matching the example is enough: Require the new W4 case, explicit **prediction**s and an **explanation**. Fresh retry: B (2,4,8), C (3,2,8), A (5,5,9), limit 8. B/C tie at 8; B wins. Limit 7 returns None.

A convenient shortcut preserves meaning: Updating best on score >= best_score makes the last tied item win, so reordering can change results. Use the explicit name comparison for ties.

Test both orders and a no-match case after the repair.

Choose support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows	Next teaching action
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral **explanation**, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Extension misconceptions

A successful run proves the conclusion: Each **stage** should have a clear **input** and **output** contract. A program can be correct at one **stage** but wrong at their connection, such as passing rejected rows into selection.

Keep acceptance and rejection counts in the report.

Matching the example is enough: Require the new W4 case, explicit **prediction**s and an **explanation**. Fresh retry: X (minutes 9, fun 2, learning 3), Y (10,3,2), Z (11,6,1), limit 10.

Accept X/Y, reject Z rating; select Y score 8 versus X 7.

A convenient shortcut preserves meaning: Calling the selector on raw data can crash or produce a result from invalid ratings. Route data through validation first. Duplicate names require a batch check even when each row individually passes; exclude every occurrence of each duplicated name, retain reasons and review the source instead of arbitrarily keeping one.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Make the lab runnable

Use the supplied main investigation for shared practice. Reserve Rank the new mystery catalogue independently for the individual first attempt. The named extension has separate tasks and feedback.

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Check these misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected main independent task

> Reveal its key after the first attempt; extension tasks have their own keys.

Selected independent question and key

Selected case: Request: mystery, at most 100 pages. Catalogue: D mystery110 illustrated; E mystery80 plain; F science80 illustrated; G mystery95 illustrated; H mystery90 illustrated.

Filter first; score 3 for mystery plus 2 for illustrations; break ties by fewer pages.

D fails length and F fails genre. E scores3, G and H score5. H wins the tie with 90 versus95 pages. Filtering is separate from scoring; this written preference rule is not proof of a trained model.

Judge reasoning and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Trace selection

Why does A win the worked example even though B is **process**ed first?

Both score 8 and qualify; A is alphabetically first under the tie rule.

W2 Handle no match

State recommend([], 10) and the worked items with limit 8.

Both return None; the caller should display no eligible option.

For the empty list, the loop has no iterations, so best stays None. With the worked durations 10 and 9 under a limit of 8, neither item passes the eligibility check, so best again stays None.

The caller must turn this result into a clear no-eligible-option message; it must not treat None as an activity name.

W3 Repair order dependence

A tied selection changes after reversing **input** order. Identify the likely missing policy and test.

An explicit **deterministic** tie rule is missing or incorrect. Apply alphabetical name order and rerun both orders.

W5 Explain the result

Write a one-sentence **explanation** of X winning without calling it objectively best.

X meets the 12-minute limit and ties for highest eligible score 9; the alphabetical tie rule selects X.

<details>
<summary>Extension questions and keys</summary>

Building the data pipeline — W1 Trace a validation

Explain why a row with fun "4" fails while fun 4 can pass.

The first is text; the schema requires an exact integer rating from 0 to 5.

Building the data pipeline — W2 Check missing fields

What does valid_item return if learning is absent?

False: get returns None and the exact integer check fails.

When learning is absent, row.get("learning") returns None. The condition type(value) is not int is true, so the helper returns False before doing the range comparison.

This is a schema rejection; it is different from a valid item that later exceeds the selection time limit.

Building the data pipeline — W3 Repair the connection

A program validates rows but still sends the original raw list to recommend. What should change?
</details>

Pass only the accepted, uniqueness-checked list. Preserve rejection information separately for reporting.

Building the data pipeline — W4 Build independently

Combine validation and the selector in this week's lesson for A (minutes 10, fun 3, learning 2), B (12,4,1), and C (8,"5",5), limit 10. State accepted rows, rejected row, selection and **explanation**.

A and B validate; C fails fun type. B exceeds the limit. A wins with score 8. Report 2 accepted of 3, one rejection and one eligible item. Verify unique names.

Building the data pipeline — W5 Handle duplicates

Two valid rows have the same name but different ratings. Should individual validation alone permit selection?

No. The batch uniqueness contract fails. Exclude every occurrence of each duplicated name and report the reasons for correction, following the frozen Week32 policy.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

A wins at 3; B=2; C=1. Show flags, formula, scores and **explanation**; declare tie policy.

Additional transfer practice

A reader requests space and at most 60 pages. Book A is space, 90 pages; B is animals, 40 pages. What should the prototype return if both conditions are required?

Reasoned practice answer

Neither matches both conditions. Return no exact match and ask whether to relax a preference; do not present A or B as a full match.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions

Criterion	0 points	1 point	2 points
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “The system must recommend something even when constraints fail.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Only A meets genre and length constraints; its score is 3. B exceeds the page limit and C fails genre. Do not claim A beat eligible competitors when none existed. Include no-match and missing-field handling in the prototype.

Explanations for the additional workbook examples

W1 Trace selection

W2 Handle no match

W3 Repair order dependence

W4 Implement independently

Use the function with X (fun 4, learning 1, minutes 12), Y (3,3,10), Z (5,5,13), limit 12. Predict the winner, reverse the list and retest. Also test limit 9.

X and Y score 9; Z is excluded. X wins alphabetically in either order. Limit 9 returns None.

W5 Explain the result

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and return

> Keep main and extension assessment evidence distinguishable.

Project record

Keep the decision, **input**, result, **explanation** and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Save the **recommender** code, contract and this full trace; test the same records in different **input** orders.

Main support and fresh assessment

Trace best and best_score row by row. Extension: return a record with name, score and reason while retaining the same selection policy.

Selected independent mystery task: D fails length, F fails subject; E scores3 and G/H5; H wins the shorter-page tie90<95. Require the filtering and score trace. Fresh book retry: requestanimal,max80pages; score2illustrated+1comic,tieID; J animal80 plaincomic, K animal70 illustratedprose, L space60 illustratedcomic. Expected J1,K2;Lexcluded;Kselected. Collect before feedback and record support used. The older X/Y/Z activity task remains additional practice with its own key.

Delayed return

Suggested return points in this level: ai-2/34, ai-2/36. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 34: Test Diverse Users. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week34: Test Diverse Users. Reuse the implemented baseline, contract, complete traces and expected **output**s when auditing cases and failure conditions.

▼ Extension support and reassessment

Connect paper **stage** cards with **input**/**output** labels. Extension: return specific reasons for missing, type and range errors without exposing personal data.

Independent W4 **evidence**: A and B validate; C fails fun type. B exceeds the limit. A wins with score 8. Report 2 accepted of 3, one rejection and one eligible item. Verify unique names.

Record whether the learner needed no help, a hint or **explanation**. Fresh retry: X (minutes 9, fun 2, learning 3), Y (10,3,2), Z (11,6,1), limit 10. Accept X/Y, reject Z rating; select Y score 8 versus X 7.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Transparent Book Recommender

This is the complete **baseline**: **input**s, scoring, filter, ranking, **explanation**, and abstain rule.

Support and stretch

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Exit reflection

Bridge to the next week

Support and fresh reassessment

Readiness and reassessment for Building the data pipeline

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative teaching checklist**

REFERENCE · SECTION 7

Reference · Alternative teaching checklist

> Use as a reminder alongside the main route.

Planning reference

Teaching target: Implement a **deterministic recommender**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Before class, solve the supported practice: State recommend([], 10) and the worked items with limit 8.

Reasoned practice answer: Both return None; the caller should display no eligible option.

Check every learner before discussion: Use the worked case for Implementing a **recommender**. Proposed **explanation**: “A successful run proves the conclusion”. Decide whether this **explanation** is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: The recommendation follows a written scoring policy. It does not learn preferences, prove learning benefit or authorize an action.

Display the chosen rule and let a person review whether an option is suitable.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken **explanation** or drawing; retain the reasoning goal.

Extension: Change D’s pages to 91, leaving every other field unchanged. Require a **prediction** and a reason before comparing. Expected result: D becomes ineligible, so A wins with score 2.

Reason: The strict excess over the 90-page maximum changes filtering before ranking.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's **explanation** and **evidence**.

▼ **Reference · Feedback and retained evidence**

REFERENCE · SECTION 8

Reference · Feedback and retained evidence

> Keep supported practice separate from independent evidence.

Record and repair

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Use the earlier worked-**explanation** links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess **explanation**, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an **explanation**. The branch-boundary and indexing lessons deliberately include failure cases.
</details>

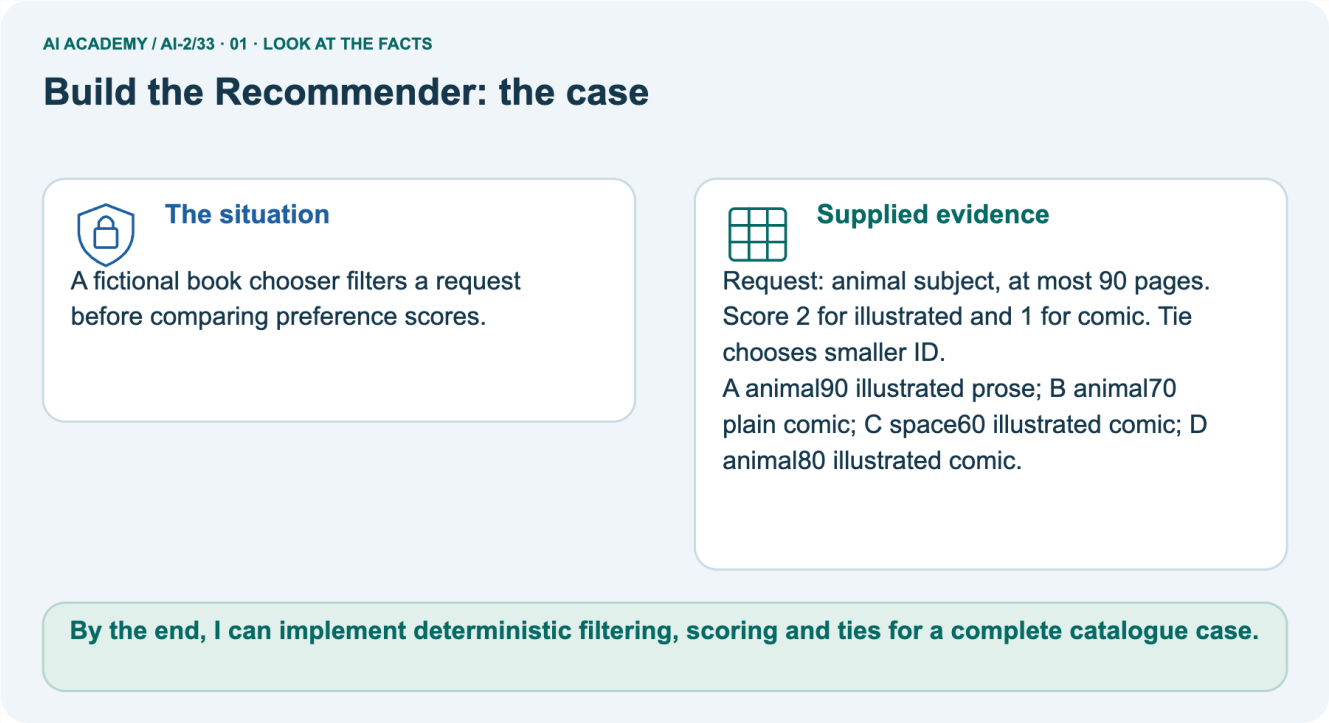
<details>
<summary>Reference · Teach with the six pictures</summary>
</details>

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose a picture for a specific explanation.

01. Build the Recommender: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

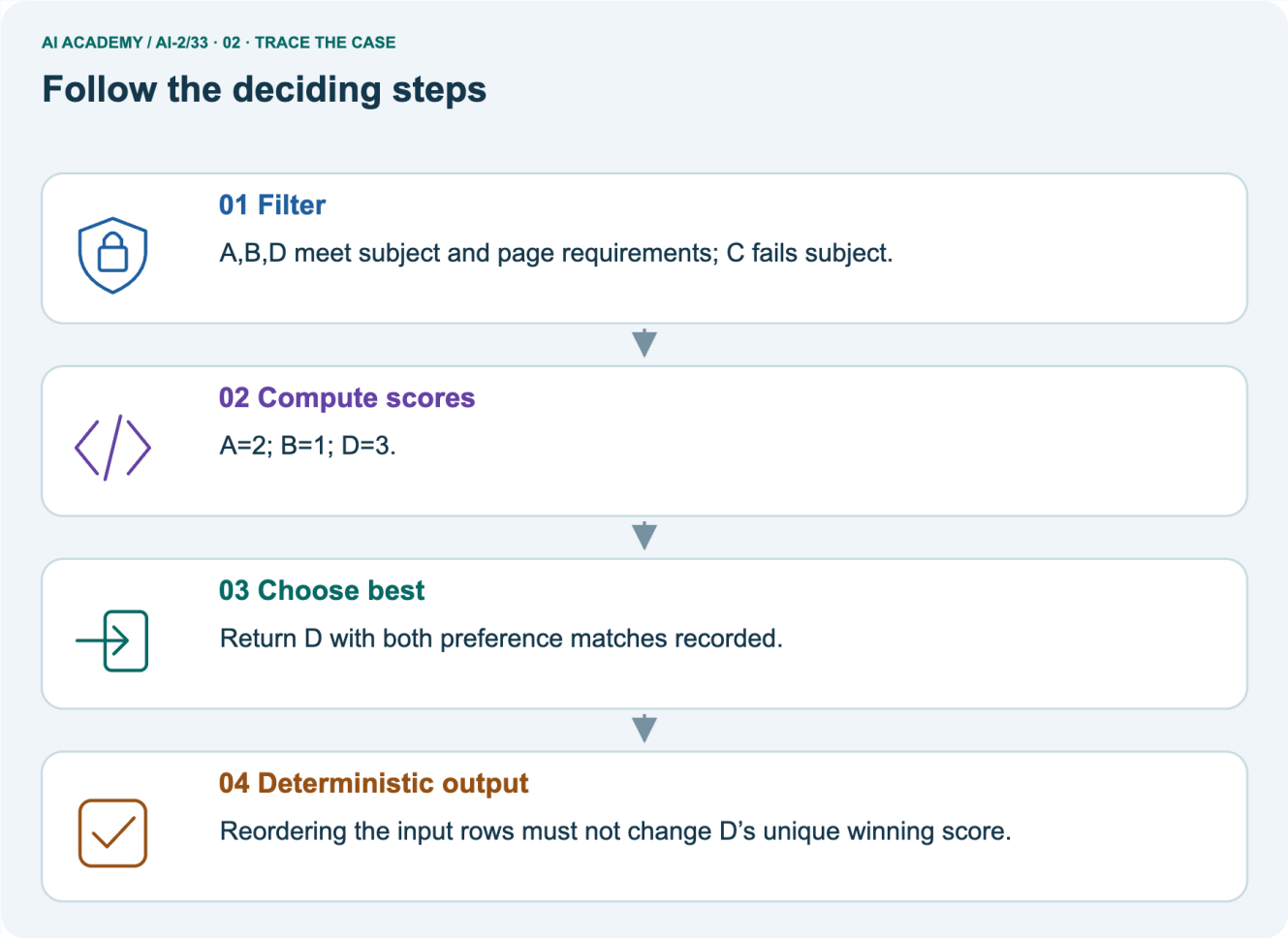
A fictional book chooser filters a request before comparing preference scores. Supplied facts: Request: animal subject, at most 90 pages. Score 2 for illustrated and 1 for comic. Tie chooses smaller ID.

A animal90 illustrated prose; B animal70 plain comic; C space60 illustrated comic; D animal80 illustrated comic.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Filter: A,B,D meet subject and page requirements; C fails subject. → Compute scores: A=2; B=1; D=3. → Choose best: Return D with both preference matches recorded. → **Deterministic output**: Reordering the **input** rows must not change D's unique winning score.

Point to each numbered **stage** and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered **stage** and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Filtering before scoring



Follow the labelled picture with the matching explanation. Open the vector diagram

Filtering before scoring. A: Pages = 90, Preference score = 2, Status = eligible. B: Pages = 70, Preference score = 1, Status = eligible. C: Pages = 60, Preference score = 3, Status = subject fails.

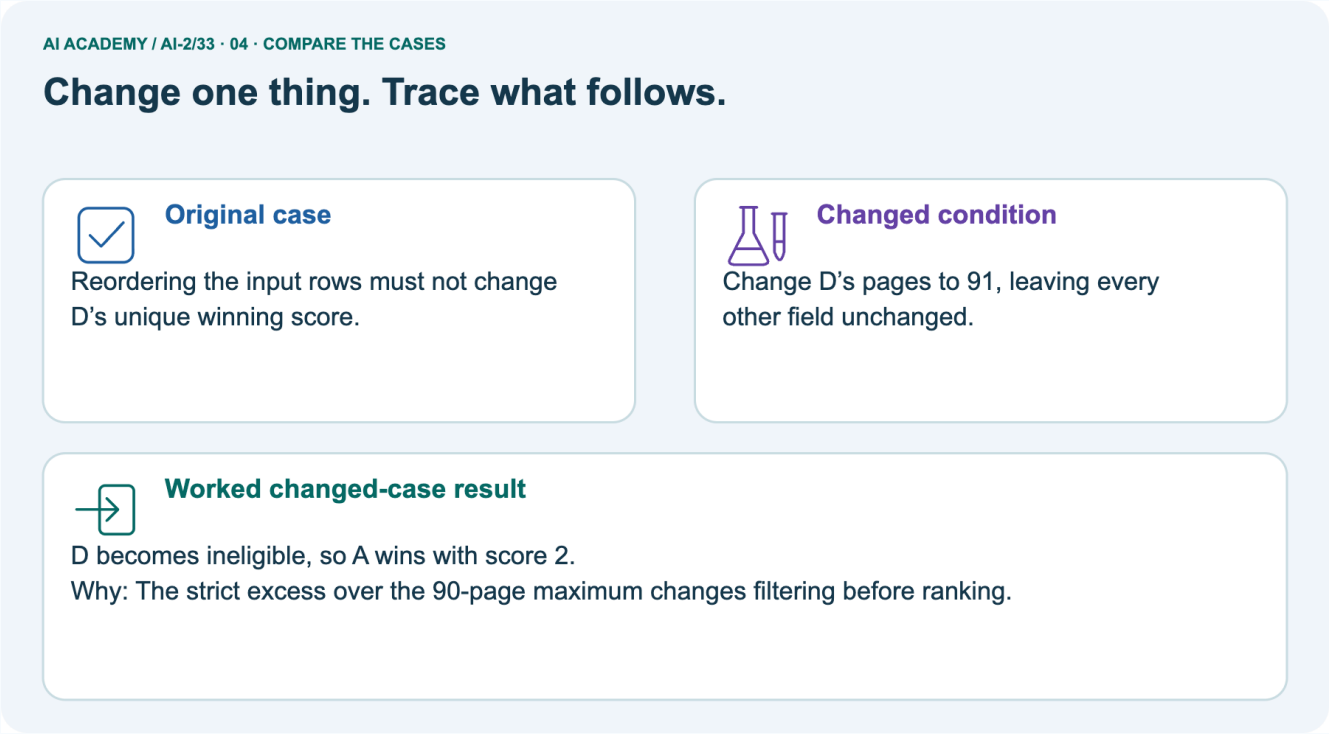
D: Pages = 80, Preference score = 3, Status = eligible.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Reordering the **input** rows must not change D's unique winning score. Changed condition: Change D's pages to 91, leaving every other field unchanged. Expected result: D becomes ineligible, so A wins with score 2.

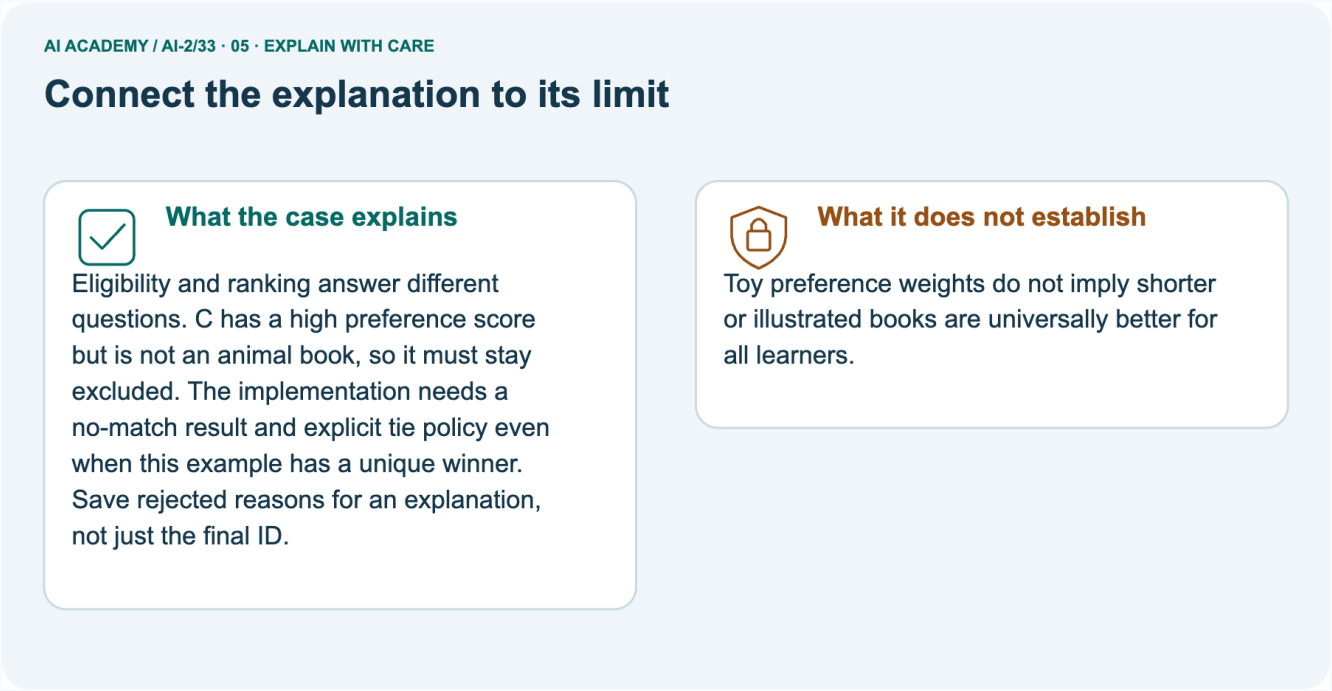
Reason: The strict excess over the 90-page maximum changes filtering before ranking.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The strict excess over the 90-page maximum changes filtering before ranking.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Eligibility and ranking answer different questions. C has a high preference score but is not an animal book, so it must stay excluded.

The implementation needs a no-match result and explicit tie policy even when this example has a unique winner. Save rejected reasons for an **explanation**, not just the final ID.

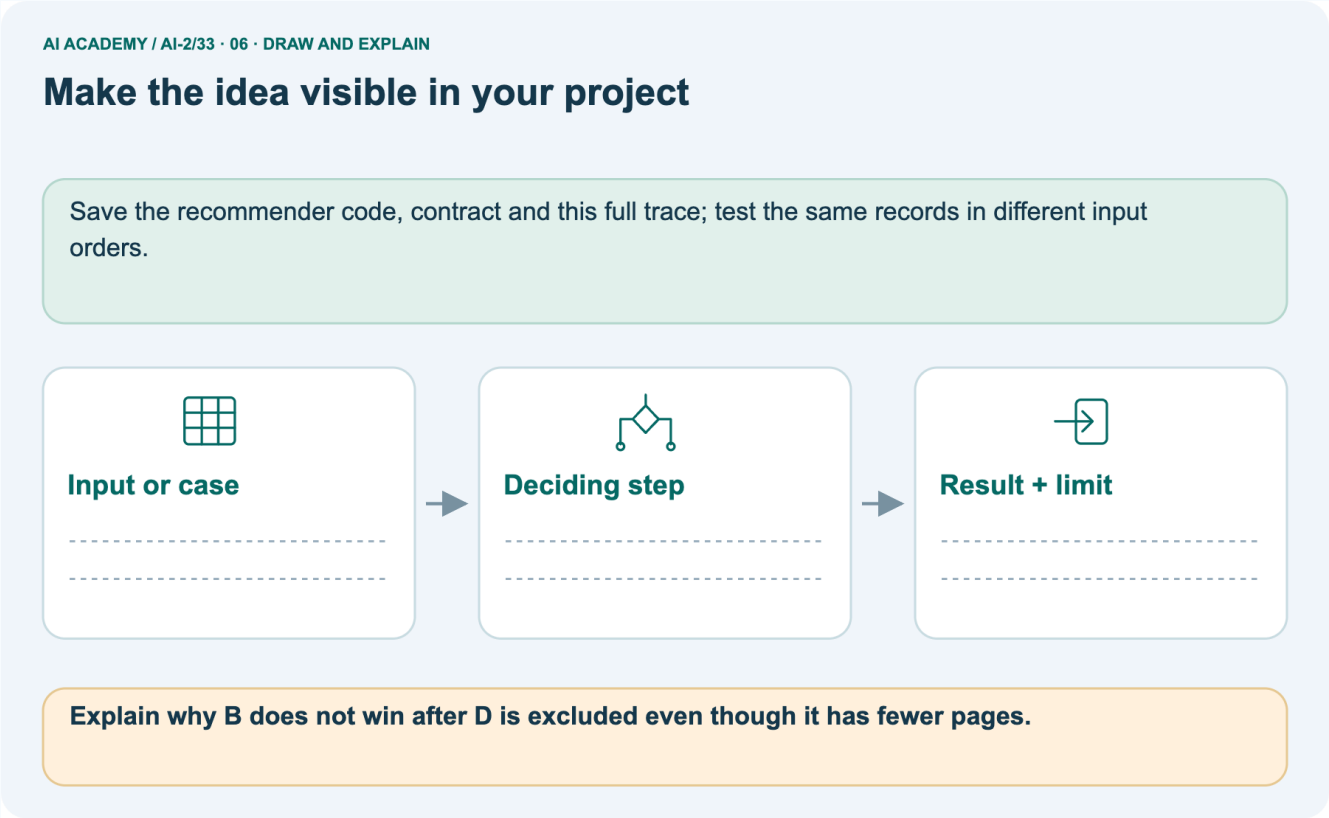
Limit: Toy preference weights do not imply shorter or illustrated books are universally better for all learners.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Toy preference weights do not imply shorter or illustrated books are universally better for all learners.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Save the **recommender** code, contract and this full trace; test the same records in different **input** orders. Draw your **input**, deciding step, result and limit.

Supported question: Explain why B does not win after D is excluded even though it has fewer pages.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Page count is an eligibility limit, not a ranking preference here. A scores 2 and B scores 1.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Build the Recommender: the case

02. Follow the deciding steps

03. Filtering before scoring

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.